## Assignment 7:

Write a C program that includes a user-defined function named binarySearch with the signature int binarySearch(int arr[], int size, int target);. The function should perform a binary search on a sorted array of integers and return the index of the target element if found, and -1 otherwise.

## Source Code:

```
#include<stdio.h>
int binarySearch(int arr[], int size, int target) {
   int low = 0, mid, high = size-1;
   while (low<=high) {
      mid = (low+high)/2;
      if (arr[mid]==target)
         return mid;
      else if (target<arr[mid])
         high = mid-1;
      else
         low = mid+1;
   }
   return -1;
}
int main() {
   int i, arr[10], n, target, result;
   printf("Enter the size of the Array : ");
   scanf("%d",&n);
   printf("Enter the Elements of the sorted array in ascending order : ");
   for (i=0;i<n;i++)
      scanf("%d",&arr[i]);
   printf("Enter the target element : ");
   scanf("%d",&target);
   result = binarySearch(arr,n,target);
   if (result==-1)
      printf("The targeted element is not found.");
   else
      printf("The targeted element is at index : %d",result);
   return 0;
}
```

## Output:

```
Enter the size of the Array :3

Enter the Elements of the sorted array in ascending order :2 5 8

Enter the target element :8

The targeted element is at index : 2
Process finished with exit code 0
```

```
Enter the size of the Array :4

Enter the Elements of the sorted array in ascending order :58 45 72 99

Enter the target element :72

The targeted element is at index : 2
Process finished with exit code 0
```